

AA411

Diagnoser Test Plan

Team Name: ROFASO

Project Name: Damage Assessment Protocol (D.A.P.)

Sub-system Name: Fault Isolator

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1 Objective of Test Plan

The goal of this plan is to test that the Diagnoser can accurately tabulate the most likely fault cases by lowest percent difference of mission data to greatest percent difference.

i Scope

- Function: Diagnoser
- Type: Code verification
- Outputs: Tabulated likely fault cases from most likely to least likely

2 Test Parameters and Variables

i Parameters

Parameters	Description	Value	Source
Mission Data	The data from the Seaglider mission	All	SG.github
Simulated digital twin (DT) Data	The data from the mission representing a functional DT	All	DT Code

ii Variables

Variable	Description	Units
Independent Variables (Control Inputs)		
100 % Nominal	Fault Case with no modifications to the Seaglider	Mission data units
50 % Wing	Fault Case with 50 percent of the wing span remaining	Mission data units
0 % Rudder	Fault Case with 0 percent of the rudder span remaining	Mission data units
Dependent Variables (State Variables)		
PitchAngle	Pitch of Seaglider	(deg)
RollAngle	Roll of Seaglider	(deg)
Heading	Orientation of Seaglider	(deg)
PitchAngleRate	Rate of change of Pitch of Seaglider	(deg/s)
RollAngleRate	Rate of change of Roll of Seaglider	(deg/s)
HeadingRate	Rate of Change of Orientation of Seaglider	(deg/s)

3 Test Matrix

Full Test Matrix:

Test ID	Mission Input	DT Input 1	DT Input 2	DT Input 3	Score 1	Score 2	Score 3	Package Configuration
P1R1	MI_{N5}	MI_{N5}	MI_{W11}	MI_{R8}	Nominal	Rudder	Wing	N5 / R8 / W11
P1R2	MI_{R8}	MI_{N5}	MI_{W11}	MI_{R8}	Rudder	Wing	Nominal	N5 / R8 / W11
P1R3	MI_{W11}	MI_{N5}	MI_{W11}	MI_{R8}	Wing	Rudder	Nominal	N5 / R8 / W11
P2R1	MI_{N5}	MI_{N5}	MI_{W12}	MI_{R8}	Nominal	Rudder	Wing	N5 / R8 / W12
P2R2	MI_{R8}	MI_{N5}	MI_{W12}	MI_{R8}	Rudder	Wing	Nominal	N5 / R8 / W12
P2R3	MI_{W12}	MI_{N5}	MI_{W12}	MI_{R8}	Wing	Rudder	Nominal	N5 / R8 / W12
P3R1	MI_{N5}	MI_{N5}	MI_{W11}	MI_{R9}	Nominal	Rudder	Wing	N5 / R9 / W11
P3R2	MI_{R9}	MI_{N5}	MI_{W11}	MI_{R9}	Rudder	Wing	Nominal	N5 / R9 / W11
P3R3	MI_{W11}	MI_{N5}	MI_{W11}	MI_{R9}	Wing	Rudder	Nominal	N5 / R9 / W11
P4R1	MI_{N5}	MI_{N5}	MI_{W12}	MI_{R9}	Nominal	Rudder	Wing	N5 / R9 / W12
P4R2	MI_{R9}	MI_{N5}	MI_{W12}	MI_{R9}	Rudder	Wing	Nominal	N5 / R9 / W12
P4R3	MI_{W12}	MI_{N5}	MI_{W12}	MI_{R9}	Wing	Rudder	Nominal	N5 / R9 / W12
P5R1	MI_{N6}	MI_{N6}	MI_{W11}	MI_{R8}	Nominal	Rudder	Wing	N6 / R8 / W11
P5R2	MI_{R8}	MI_{N6}	MI_{W11}	MI_{R8}	Rudder	Wing	Nominal	N6 / R8 / W11
P5R3	MI_{W11}	MI_{N6}	MI_{W11}	MI_{R8}	Wing	Rudder	Nominal	N6 / R8 / W11
P6R1	MI_{N6}	MI_{N6}	MI_{W12}	MI_{R8}	Nominal	Rudder	Wing	N6 / R8 / W12
P6R2	MI_{R8}	MI_{N6}	MI_{W12}	MI_{R8}	Rudder	Wing	Nominal	N6 / R8 / W12
P6R3	MI_{W12}	MI_{N6}	MI_{W12}	MI_{R8}	Wing	Rudder	Nominal	N6 / R8 / W12
P7R1	MI_{N6}	MI_{N6}	MI_{W11}	MI_{R9}	Nominal	Rudder	Wing	N6 / R9 / W11
P7R2	MI_{R9}	MI_{N6}	MI_{W11}	MI_{R9}	Rudder	Wing	Nominal	N6 / R9 / W11
P7R3	MI_{W11}	MI_{N6}	MI_{W11}	MI_{R9}	Wing	Rudder	Nominal	N6 / R9 / W11
P8R1	MI_{N6}	MI_{N6}	MI_{W12}	MI_{R9}	Nominal	Rudder	Wing	N6 / R9 / W12
P8R2	MI_{R9}	MI_{N6}	MI_{W12}	MI_{R9}	Rudder	Wing	Nominal	N6 / R9 / W12
P8R3	MI_{W12}	MI_{N6}	MI_{W12}	MI_{R9}	Wing	Rudder	Nominal	N6 / R9 / W12

Table 1: Full Diagnoser Validation Test Matrix

Test ID	Mission Input	DT Input 1	DT Input 2	DT Input 3	Score 1	Score 2	Score 3	Package Configuration
P1R1	MI_{N5}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Nominal	Rudder	Wing	N5 / R8 / W11
P1R2	MI_{R8}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Rudder	Wing	Nominal	N5 / R8 / W11
P1R3	MI_{W11}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Wing	Rudder	Nominal	N5 / R8 / W11
P2R1	MI_{N5}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Nominal	Rudder	Wing	N5 / R8 / W12
P2R2	MI_{R8}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Rudder	Wing	Nominal	N5 / R8 / W12
P2R3	MI_{W12}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Wing	Rudder	Nominal	N5 / R8 / W12
P3R1	MI_{N5}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Nominal	Rudder	Wing	N5 / R9 / W11
P3R2	MI_{R9}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Rudder	Wing	Nominal	N5 / R9 / W11
P3R3	MI_{W11}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Wing	Rudder	Nominal	N5 / R9 / W11
P4R1	MI_{N5}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Nominal	Rudder	Wing	N5 / R9 / W12
P4R2	MI_{R9}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Rudder	Wing	Nominal	N5 / R9 / W12
P4R3	MI_{W12}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Wing	Rudder	Nominal	N5 / R9 / W12
P5R1	MI_{N6}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Nominal	Rudder	Wing	N6 / R8 / W11
P5R2	MI_{R8}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Rudder	Wing	Nominal	N6 / R8 / W11
P5R3	MI_{W11}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Wing	Rudder	Nominal	N6 / R8 / W11
P6R1	MI_{N6}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Nominal	Rudder	Wing	N6 / R8 / W12
P6R2	MI_{R8}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Rudder	Wing	Nominal	N6 / R8 / W12
P6R3	MI_{W12}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Wing	Rudder	Nominal	N6 / R8 / W12
P7R1	MI_{N6}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Nominal	Rudder	Wing	N6 / R9 / W11
P7R2	MI_{R9}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Rudder	Wing	Nominal	N6 / R9 / W11
P7R3	MI_{W11}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Wing	Rudder	Nominal	N6 / R9 / W11
P8R1	MI_{N6}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Nominal	Rudder	Wing	N6 / R9 / W12
P8R2	MI_{R9}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Rudder	Wing	Nominal	N6 / R9 / W12
P8R3	MI_{W12}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Wing	Rudder	Nominal	N6 / R9 / W12

Table 2: Full Diagnoser Noise Test Matrix

4 Traceability

Requirement F-I1 is verified by test cases P1R1 through P8R3. True/False? _____

Test packages:

$MI_{N5} = \text{Nominal Dive 5}$

$MI_{N6} = \text{Nominal Dive 6}$

$MI_{R8} = 0\% \text{ Rudder Dive 8}$

$MI_{R9} = 0\% \text{ Rudder Dive 9}$

$MI_{W11} = 50\% \text{ Wing Dive 11}$

$MI_{W12} = 50\% \text{ Wing Dive 12}$

Test Runs per Package:

$MI_N = \text{Nominal Mission Input}$

$MI_{50W} = 50\% \text{ Wing Fault Mission Input}$

$MI_{0R} = 0\% \text{ Rudder Fault Mission Input}$

Run 1: $MI_{\text{input}} = MI_N$; $DT_{\text{input}} = \{MI_N, MI_{50W}, MI_{0R}\}$

Run 2: $MI_{\text{input}} = MI_{0R}$; $DT_{\text{input}} = \{MI_N, MI_{50W}, MI_{0R}\}$

Package	Nominal Dataset	Rudder Dataset	Wing Dataset
1	MI_{N5}	MI_{R8}	MI_{W11}
2	MI_{N5}	MI_{R8}	MI_{W12}
3	MI_{N5}	MI_{R9}	MI_{W11}
4	MI_{N5}	MI_{R9}	MI_{W12}
5	MI_{N6}	MI_{R8}	MI_{W11}
6	MI_{N6}	MI_{R8}	MI_{W12}
7	MI_{N6}	MI_{R9}	MI_{W11}
8	MI_{N6}	MI_{R9}	MI_{W12}

Table 3: Diagnoser Validation Test Packages

Test Run	Mission Input	Simulated DT Input 1	Simulated DT Input 2	Simulated DT Input 3	Expected Best Match
1	MI Nominal	MI Nominal	MI 50W	MI 0R	Nominal
2	MI 50R	MI Nominal	MI 50W	MI 0R	0R
3	MI 50W	MI Nominal	MI 50W	MI 0R	50W

Table 4: Diagnoser Validation Test Runs per Package

Run 3: $MI_{\text{input}} = MI_{50W}$; $DT_{\text{input}} = \{MI_N, MI_{50W}, MI_{0R}\}$